ArtComputer

Functions User Manual

Automated Test with the Robot

Functions User Manual

Contents

Foreword

This user manual will give you detail on the functions available in the tests and in the rules.

Tests versus Rules

In the Rules, you can use all the available functions for the tests.
However, for the rules, the name of the function is prefixed by # and ended by : (E.g.: #click:)
The parameters are separated by a comma (E.g: #click: @OPSYS_Listbox, 5, 2)

Naming convention

- Dictionary starts always with @
- Dataset starts always with #
- Rule starts always with #
- Variable start always with $ (but in the case of a rule, it must be §)
- Naming convention can be all in uppercase or lowercase: @URL_ACCEPTANCE
- Naming convention can be a mix @URL_Acceptance
- Naming convention can be with spaces or not: @URL_Environment Acceptance
- Functions name are case sensitive

Functions to start

Function: url

Objectives
Method to browse a website

Parameter(s)

URL	Link to the webpage to open (can be a word in the dictionary)

Example(s):

url	@URL_OPSYS_ACC EUROPA	Start the application in acceptance
url	https://webgate.acceptance.ec.europa.eu/mwp/home?1fa	

Function: geturl

Objectives
Method to get the current url from the browser

Parameter(s)

Variable	The name of the variable to store the current url (starting with $)

Example(s):

geturl	$currentURL	Store the current url into the variable $currentURL

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #geturl: §currentURL

Function: getUrlTitle

Objectives

Method to get the title of the current web page

Parameter(s)

Variable	The name of the variable to store the title (starting with $)

Example(s):

getUrlTitle	$Title	Store the title of the current web page into the variable $Title

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #getUrlTitle: §Title

Function: switchTab

Objectives

Method to switch to a specific tab on the browser

Parameter(s)

Tab position	tab position starting by 1 (0 for the last one)

Example(s):

switchTab	0	Switch to the last tab
switchTab	2	Switch to the second tab

Function: openNewTab

Objectives

Method to open new tab on the browser (the tab will become the active one)
Note: with playwright the method is equivalent to open a new window!

Parameter(s)

Url	Url text or a word in the dictionary or a variable name starting with $

Example(s):

openNewTab	@URL_javadoc	Create a new tab on the browser with the link

Function: clickNewTab

Objectives

Click to open a new tab in the browser.
Note: Contrary to the function openNewTab, the clickNewTab open a real tab in the browser

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI

Example(s):

clickNewTab	@link_javadoc_button	Click on the button to open a new tab on the browser with the link

Function: closeTab

Objectives

Close the current tab and back to the first one.

Example(s):

closeTab		Close the current tab and back to the first one.

Function: loginUser

Objectives

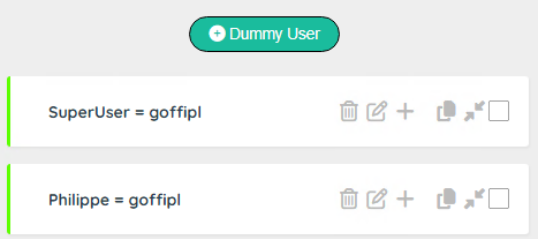
Method to login the user to an application.

Parameter(s)

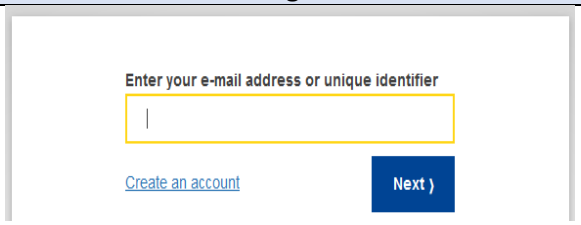
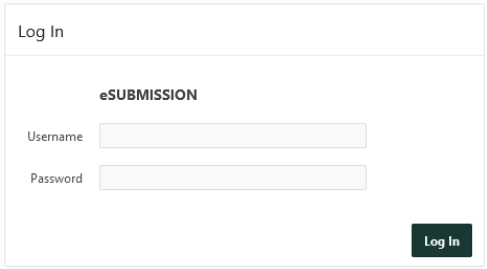
Dummy user	The dummy user must be defined in the table (see: Tester User Manual)
User tag	Enter the xpath (or dictionary word) to access the user field
Submit tag	[Optional] Enter the xpath (or dictionary word) to access the submit button

Example(s):

The dummy user must be defined in the dummy user entity



loginUser	Philippe, @APP_tagLogin, @APP_tagSubmitLogin	Login the user to a screen with a submit button.
loginUser	Philippe, @APP_tagLogin	Login the user to a screen without a submit button.

Screen with a submit tag	Screen without a submit tag
	

Function: loginPassword

Objectives

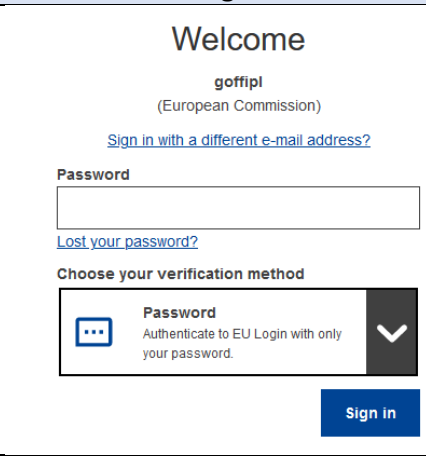
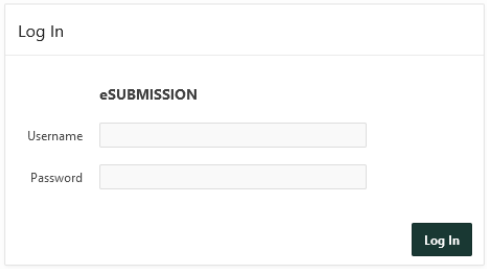
Method to key the password to an application.
For security reason, the password is decrypted (if necessary) by the server with the information contain in the dummy entity.

Parameter(s)

Dummy user	The dummy user must be defined in the table (see: Tester User Manual)
Password tag	Enter the xpath (or dictionary word) to access the password field
Submit tag	Enter the xpath (or dictionary word) to access the submit button

Example(s):

loginUser	Philippe, @APP_tagPassword, @APP_tagSubmitPassword	Key the password to a login screen.

Screen with a submit tag	Screen without a submit tag
	

Function: dummyExtraInfo

Objectives

Method to get the extra info stored in the dummy user entity.
This information can be useful to complete a login.
It can be useful with the <ME> parameter to get information of the connected person

Parameter(s)

Dummy user	The dummy user must be defined in the table (see: Tester User Manual)
Variable	The name of the variable to store information (starting with $)

Example(s):

dummyExtraInfo	Philippe, $PhoneName	Get the phone name of the dummy user.
dummyExtraInfo	<ME>, $PhoneName	Get the phone name of the connected user.

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #dummyExtraInfo: Philippe, §PhoneName

Function: dummyLogin

Objectives

Method to get the login info stored in the dummy user entity.
This information can be useful with the <ME> parameter to get the login of the connected person.

Parameter(s)

Dummy user	The dummy user must be defined in the table (see: Tester User Manual)
Variable	The name of the variable to store information (starting with $)

Example(s):

dummyLogin	<ME>, $MyLogin	Get the login of the connected user.

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #dummyLogin: <ME>, §MyLogin

Function: speak

Objectives

Method to text-to-speech a message

Parameter(s)

Text	Text to say (can be variable starting with $)

Example(s):

speak	Hello dear tester	Say a sentence

Function: debug

Objectives

Method to display more or less message to the console

Parameter(s)

Debug level	0: No Debug, 1: Important info, 2: Full detail

Example(s):

debug	0	No debug message sent to the console

Basic functions

Function: detectGUI

Objectives

Method to detect the signature of an element based on generic patterns.
Patterns are generated by the AI Robot (see specific user documentation on AI Robot).
If detectGUI is successful, the variable **$GUI** will contain the xpath to access the element

Parameter(s)

Element	Select a selector (Button, Input filed…). selector depends on the project (stored in the entity: Selector in the AI Robot)
Criteria	Enter the criteria (generally the label)
Position	Enter the position/occurrence (1 by default), $$ for last record or $$-<position>
Stop on Error	[Optional]: Stop on error (otherwise a warning is sent)

Example(s):

detectGUI	Button, Save, 1 Button, Save, 1, 0	Searching for the button ‘Save’. If not found, send a warning and continue the tests
detectGUI	Button, Save, 1, 1	Searching for the button ‘Save’. If not found, stop all the tests
detectGUI	Button, Save, $$	Searching for the last button ‘Save’

Function: pause

Objectives

Method to wait a few seconds before continuing the next step.

Parameter(s)

Delay	Select a selector (Button, Input filed…). selector depends on the project (stored in the entity: Selector in the AI Robot)

Example(s):

pause	3	Wait 3 seconds

Function: waitFor

Objectives

Method to wait for the refresh of an element (to be visible).

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait	Waiting time in second(s) (default 5 sec)
Continue	What to do if the element is not ready after the waiting time: 1: Stop all the tests, 0: Continue even with an error, 2: Skip the IT

Example(s):

waitFor	@APP_Save, 5, 1	Wait 5 seconds for the button Save to be visible. If it is not the case, stop all the tests

Function: waitForNot

Objectives

Method to wait for the element to disappear (reverse of waitFor)

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait	Waiting time in second(s) (default 5 sec)
Continue	What to do if the element is still there after the waiting time: 1: Stop all the tests, 0: Continue even with an error, 2: Skip the IT

Example(s):

waitForNot	@APP_In Progress, 15, 1	Wait 15 seconds for the text “in progress…” disappears. If it is not the case, stop all the tests

Function: click

Objectives
Method to click on an element.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait after	Waiting time in second(s) after the click
Focus	1: set the focus on the element, 0: no focus on the element

Example(s):

click	@APP_Save, 3, 1	Set the focus on the button 'Save', click on it and wait 3 seconds

Function: doubleClick

Objectives
Method to double click on an element.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait after	Waiting time in second(s) after the double click

Example(s):

doubleClick	@APP_Save, 3	Double click on the button 'Save' and wait 3 seconds

Function: JSclick

Objectives

JavaScript method to click on an element this method is more brutal force).
Can be used, if the click() is not working due to an invalid webpage status (stale)

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait after	Waiting time in second(s) after the click

Example(s):

JSclick	@APP_Save, 3	Click on the button 'Save' (CSS/Xpath in the dictionary) and wait 3 seconds

Function: enable

Objectives

JavaScript method to enable an element (by removing disabled attributes)

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI

Example(s):

enable	@APP_Save	Make the button save enabled

Function: removeAttribute

Objectives

JavaScript method to remove attribute of an element.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Attribute	Attribute to be removed

Example(s):

removeAttribute	@APP_Save, color	Remove the attribute: color

Function: setAttribute

Objectives

JavaScript method to remove attribute of an element.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Attribute	Attribute to assign
Value	Value of the attribute

Example(s):

setAttribute	@APP_Save, color, blue	set the attribute color: blue

Function: readAttribute

Objectives

JavaScript method to get the value of an attribute of an element.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Attribute	Attribute to read
Variable	The name of the variable to store information (starting with $)

Example(s):

setAttribute	@APP_Save, tagElement, $Tag	get the attribute tag into the variable $Tag

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #setAttribute: @APP_Save, tagElement, §Tag

Function: setFocus

Objectives

JavaScript method to get the focus on an element.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait after	Waiting time in second(s) after the click

Example(s):

setFocus	@APP_Save, 2	Set the focus on the button save

Function: JSinput

Objectives

JavaScript method to input a value into an element.
Can be used, if the setValue() is not working due to an invalid webpage status (stale) - this method is a brutal force.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Value	Value to key in the field

Example(s):

JSinput	@APP_Name, Phil	Key 'Phil' into the field 'Name'

Function: keyboard

Objectives

JavaScript method to key a value to simulate the keyboard.

Parameter(s)

Value	Value to key in the field

Example(s):

JSinput	Phil	Key 'Phil' into the field 'Name'

Function: pressEnter

Objectives
JavaScript method to send an Enter key

Parameter(s)

Example(s):

pressEnter		Sent an 'Enter' key

Function: pressEscape

Objectives
JavaScript method to send an Escape key

Parameter(s)

Example(s):

pressEscape		Sent an 'Escape' key

Function: pressTab

Objectives
JavaScript method to send an tab key

Parameter(s)

Number	Number of tab to send

Example(s):

pressTab	2	Sent 2 'tab' key

Function: acceptPopup

Objectives

JavaScript method to accept a JavaScript popup

Parameter(s)

Example(s):

acceptPopup		Acknowledge the popup

Function: cancelPopup

Objectives

JavaScript method to reject a JavaScript popup

Parameter(s)

Example(s):

cancelPopup		Cancel the popup

Function: popupKeys

Objectives

Execute a powershell process to sent keys to a windows popup.
For instance: sent the keys "{TAB}{TAB}{TAB}{TAB}{TAB}{ENTER}" to the certificate windows

Parameter(s)

windowTitle	Title of the popup windows (you can use the function: showAllPopups() to identify the open popup windows)
sendkeys	A string with the keys to send (E.g.: {ENTER})

Example(s):

popupKeys	My Certificate, {TAB} {TAB} {ENTER}	Send keys to the popup “My Certificate”

Function: showAllPopups

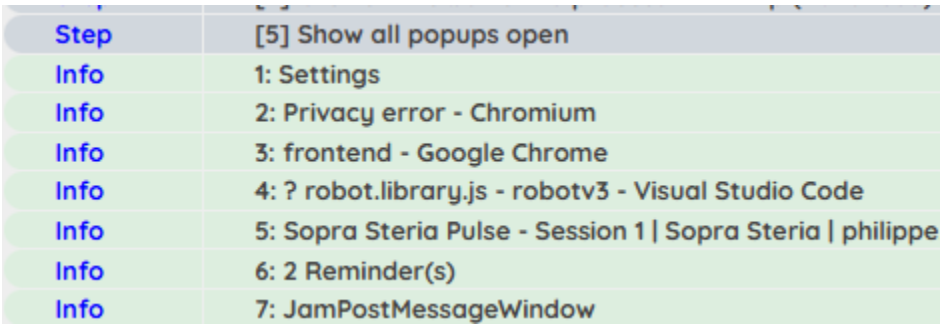
Objectives

Execute a powershell process to display all the popup titles

Parameter(s)

Example(s):

showAllPopups		Show all popup windows



Function: rule

Objectives
Method to call a rule

Parameter(s)

Rule	Name of the rule
Parameter 1	Parameter 1 for the rule (will be represented by the variable $P1)
Parameter 2	Parameter 2 for the rule (will be represented by the variable $P2)

Example(s):

Rule	Login ECAS, $DummyUser	Rule to login with a dummy user

Function: countElement

Objectives
Method to call a rule

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	The name of the variable to store information (starting with $)

Example(s):

countElement	@APP_section, $NbSection	Count the number of sections on the screen

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #countElement: @APP_section, §NbSection

Function: check

Objectives

Method to check a checkbox (if necessary)

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait after	Waiting time in second(s) after the check

Example(s):

check	@APP_approve, 5	Check the approve checkbox

Function: uncheck

Objectives

Method to uncheck a checkbox (if necessary)

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Wait after	Waiting time in second(s) after the uncheck

Example(s):

uncheck	@APP_approve, 5	Uncheck the approve checkbox

Function: message

Objectives

Method to write a message in the log file

Parameter(s)

Message	Message to display in the log file
Category	Info, Message, Warning or Error

Example(s):

message	Test successful, Info	Write a message

Function: uploadFile

Objectives

Method to upload a file from the repository uploads of your project.
The repository is managed by the Administrator.

Note: Prior to use the function, the ADMIN must store the document in the Upload section of the project.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
File	Name of the file (without a path)

Example(s):

uploadFile	@APP_Upload, price.pdf	Upload the file price.pdf

Function: ask

Objectives

Method to display a popup window to invite the user to enter a value

Parameter(s)

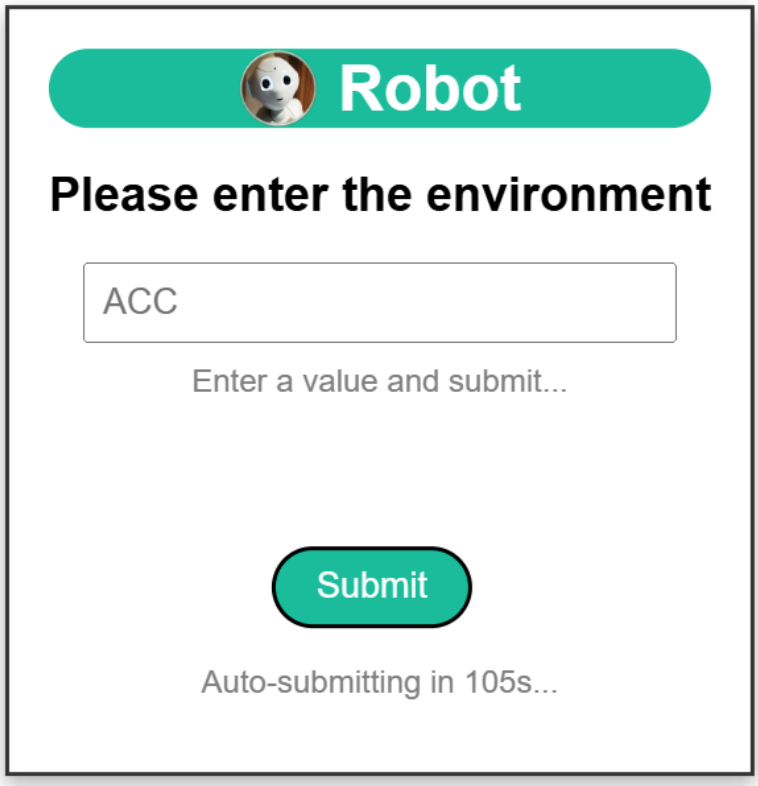
Message	The message to display to the user
Default value	[Optional] Default value
Variable	Name of the variable to store the value (by default $Ask)
Timeout	Timeout in seconds (default 30 seconds)

Example(s):

ask	Please enter the environment, ACC, $Env, 120	Ask the user to provide the name of the environment. After 120 seconds the variable $Env is filled with ACC

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #ask: Enter the environment, ACC, §Env, 20



Function: email

Objectives

Method to send an email with attachment(s) (optional).
The originator of the message is managed by the Administrator and is stored in the project parameters:

> Email Host - smtpmail.cec.eu.int:25

> Email From - automated-notifications@nomail.ec.europa.eu (AutoTest)

Parameter(s)

Email To	Recipient (comma separated for multiple people)
Subject	subject of the message
Body	body of the message (keywords: <BLUE><RED>..<BOLD><ITALIC><NORMAL> Body can contain html tag (E.g.: <table>, <body>, <tr>, <td>….)
Attachment	[optional] Full path name of the attachment(s) - use ';' as a separator

Example(s):

email	$To, $Subject, $Body, $Attachment	Send an email

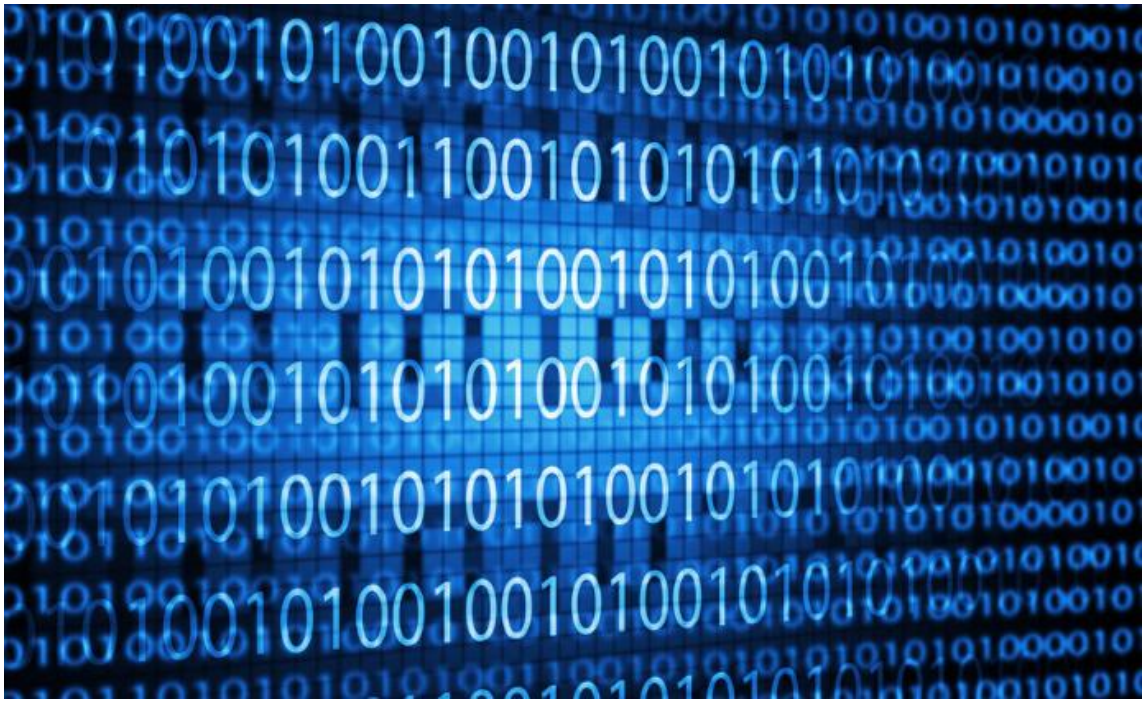
Example of body for a sanity check with an error in one of the environment. <BOLD><RED>Error detected in $Environment<NORMAL><NORMAL>

Note: a tag <NORMAL> must be added after the tags <BLUE><RED>..<BOLD><ITALIC>

In order to use the function, the ADMIN must create two parameters at the project level

Parameter name	Example
Email Host	smtpmail.cec.eu.int:25
Email From	automated-notifications@nomail.ec.europa.eu (AutoTest)

References and Data



Function: getReference

Objectives

Method to get a reference by Code. The reference is used by the Robot to exchange (read/write) data between the scenarios.

Parameter(s)

Code	Code of the reference
Variable	Name of the variable to store the value (starting with $)

Example(s):

getReference	Dataset, $Dataset	Get the value of the dataset in the reference

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #getReference: Dataset, §Dataset

Function: setReference

Objectives

Method to get a reference. The reference is used by the Robot to exchange (read/write) data between the scenarios.

Parameter(s)

Code	Code of the reference
Value	Value of the reference (can be a variable starting with $)
Comment	[]Optional If Comment is empty, the value will not be overridden

Example(s):

getReference	Dataset, $Dataset	Get the value of the dataset in the reference

Function: setVariable

Objectives

Method to set a variable.

Parameter(s)

Variable	Name of the variable (starting with $)
Value	Enter a value or <EMPTY> or an expression (must start with =)

Example(s):

setVariable	$Test, A simple test	Set a text into the variable
setVariable	$Test, = 1 + 1	Set 2 into the variable
setVariable	$Test, <EMPTY>	Reset the variable
setVariable	$Test, <TODAY>	Use a keyword to get the current date

Note 1: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #setVariable: §Test, A simple test

Note 2: Special keywords are:

With **nn** as a numeric value

<TODAY>, <TODAY+**nn**>, <TODAY-**nn**> to get the current date + or – days(s) – Format: DD/MM/YYY
<NOW>, <NOW+**nn**>, <NOW-**nn**> to get the current date + or – day(s) - Format: DD/MM/YYYY HH:mm
<YEAR>>, <YEAR+**nn**>, <YEAR-**nn**> to get the current year + or – year(s) – Format: YYYY
<MONTH>>, <MONTH+**nn**>, <MONTH-**nn**> to get the current month + or – month(s) – Format: MM
<DAY>>, <DAY+**nn**>, <DAY-**nn**> to get the current day + or – days(s) – Format: DD
<HOURS>, <HOUR+**nn**>, <HOUR-**nn**> to get the current hour + or – hour(s) – Format: HH
<SEQUENCE> to get a unique number – Format: YYYYMMDD_hmmss

Manipulation on a text

As you can use standard JavaScript in the expression of a variable (expression starts by =),
You can very easily manipulate a text:
In this example, we assume that the variable $Message contains “hello your ID is 1234567890”

Return the last 10 characters	$Test, =$Message.slice(-10)	1234567890
Return the first 5 characters	$Test, = $Message.slice(0, 5)	Hello
Return the text inside out starting from the 12rd character, with a length of 2	$Test, =$Message.substr(11, 2) As the string starts at 0, 12rd char is at the position 11	ID
Return true or false if a short text is contained within a long string	$Test, =$Message.includes(‘hello’)	True
Replace a part of the text	$Test, =$Message.replace (‘hello’, ‘Hi,’)	Hi, your ID is 1234567890
Replace all by ‘’	$Test, =$Message.replace(/<br\s*\/?>/gi, '')	All the tag will be replaced (if any)
Remove leading and trailing spaces from a word	$Test, =$Message.trim()	

Function: getData

Objectives

Method to get a data by its code. Data are store in a dataset and are managed by the Tester.

Parameter(s)

Code	Code of the data Code of the data (format: #<dataset>_<data>)
Variable	Name of the variable to store the value (starting with $)

Example(s):

getData	#Data_Dataset, $Dataset	Get a value from the dataset

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #getData: #DATA_DirectLink, §DirectLink

Function: setData

Objectives

Method to set a data with a value. Data are store in a dataset and are managed by the Tester.

Parameter(s)

Code	Code of the data Code of the data (format: #<dataset>_<data>)
Value	Name of the variable to store the value (starting with $)
Comment	Comment for the data

Example(s):

setData	#Data_Dataset, SEA-2023, Contract SEA-2023	Set a value in the dataset

Condition

Function: stopTest

Objectives

Method to stop all the tests if a condition is true.

Parameter(s)

Condition	Any valid JavaScript expression that returns true or false (or a variable)
Message	Message to display when the condition is true

Example(s):

stopTest	$Error == 1, Error detected stop the tests	Error detection

Note: Condition is a JavaScript expression so: equal is ==, not equal is !=

Function: skipDescribe

Objectives

Method to skip the Describe section if the expression is true.

Parameter(s)

Condition	Any valid JavaScript expression that returns true or false (or a variable)
Message	Message to display when the condition is true

Example(s):

skipDescribe	$Action != 'Document', Skip the document section	Skip a Describe section

Note: Condition is a JavaScript expression so: equal is ==, not equal is !=

Function: skipIt

Objectives

Method to skip the IT section if the expression is true.

Parameter(s)

Condition	Any valid JavaScript expression that returns true or false (or a variable)
Message	Message to display when the condition is true

Example(s):

skipIt	$Exits == 0, Skip the test, no field detected !	Skip a IT section

Note: Condition is a JavaScript expression

Function: isCheck

Objectives

Method to detect if an element is checked. 1: if element is checked, otherwise 0

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result: 1 or 0 (starting with $)
Wait for element	Waiting time in second(s) for the element to be ready

Example(s):

isCheck	@APP_checkbox, $Agree, 5	Check if the checkbox is checked

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #isCheck: @APP_checkbox, 5, §Agree

Function: isExist

Objectives

Method to detect if an element exists. 1: if element exists, otherwise 0

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result: 1 or 0 (starting with $)
Wait for element	Waiting time in second(s) for the element to be ready

Example(s):

isExist	@APP_checkbox, $Exist, 5	Check if the checkbox exists

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #isExist: @APP_checkbox, 5, §Exist

Function: isEnable

Objectives

Method to detect if an element is enabled. 1: if element is enabled, otherwise 0

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result: 1 or 0 (starting with $)
Wait for element	Waiting time in second(s) for the element to be ready

Example(s):

isEnable	@APP_checkbox, $Enabled, 5	Check if the checkbox is enabled

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #isEnable: @APP_checkbox, 5, §Enabled

Function: isVisible

Objectives

Method to detect if an element is visible. 1: if element is visible, otherwise 0

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result: 1 or 0 (starting with $)
Wait for element	Waiting time in second(s) for the element to be ready

Example(s):

isVisible	@APP_checkbox, $Visible, 5	Check if the checkbox is visible

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #isVisible: @APP_checkbox, 5, §Visible

Functions to manage an element

Function: switchToFrame

Objectives

Method to get a value from a field.

Note: the Robot is able to manage automatically the frame and the iFrame. This function is there only in case you need to perform a special operation.

Parameter(s)

Frame	Frame 0 is the default one

Example(s):

switchToFrame	1	Switch to frame 1

Function: getValue

Objectives

Method to get a value from a field.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result (starting with $)

Example(s):

getValue	@APP_Name, $Name	Get the value of the field Name

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #getValue: @APP_Name, §Name

Function: setValue

Objectives

Method to set a value into a field.

Parameter(s)

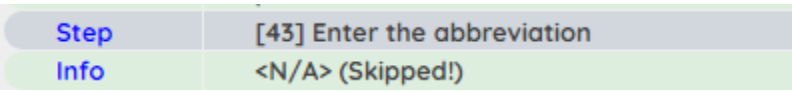
Name	Xpath or dictionary word (starting with @) of the element or $GUI
Value	Value to key (by default closed by TAB), or use <ENTER> or a variable
Delay	[Optional] Delay in second(s) before the <TAB> or <ENTER> or after keying the value – Very useful, when you have to wait for the construction of a list

Example(s):

setValue	@APP_Name, $Name	set the value in the field Name
setValue	@APP_Decision, $Decision<TAB>, 3	Enter a decision, wait for 3 sec and key a Tab

If the value is <N/A> or <EMPTY> the function will not be executed but will return with the status success.

In the logfile, the info will be <N/A> or <EMPTY> (Skipped!)



Function: select

Objectives

Method to select a value from a list.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Value	Value to key or a variable or a position (@<position>)
Wait after	Waiting time in second(s) after the click (default 2 sec)

Example(s):

select	@APP_Country, $Country, 3	Select a country

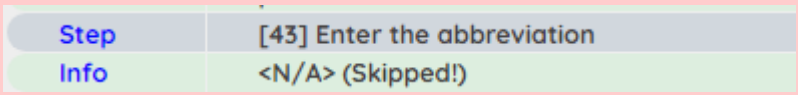
Note 1: Only the standard html <select><option> is recognized.

OBSOLETE

Note 2: Value is by default searched with a contains (approximate matching).
For compliance reason, you can use <*> but it has no impact!
To force an exact match: use = as the first character - Example: =Dupond
To get a specific option: use @<position> - Example @2 to get the second option
<Aa> is not a valid option. Value is always case sensitive.

Note 3: $Value contains the item selected from the list (useful when using position: E.g. @1)

If the value is <N/A> or <EMPTY> the function will not be executed but will return with the status success.
In the logfile, the info will be <N/A> or <EMPTY> (Skipped!)



Function: selectCount

Objectives

Method to count the number of values in a list.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result: 1 or 0 (starting with $)

Example(s):

selectCount	@APP_Country, $Countries	Count the number of countries

Note 1: Only the standard html <select><option> is recognized.

Image Functions

Function: printscreen

Objectives

Method to take a print screen and store the image on a slot (up to 5 slots available)

Parameter(s)

slot	From 1 to 5
Capture mode	**0**: Normal view, **1**: Full page

Example(s):

printscreen	1	0	Take a normal print screen and store it in the slot 1
printscreen	1	1	Take a full print screen and store it in the slot 1

Function: imageBaseline

Objectives

Method to take a print screen and store it as the baseline (to create or refresh the baseline).

Parameter(s)

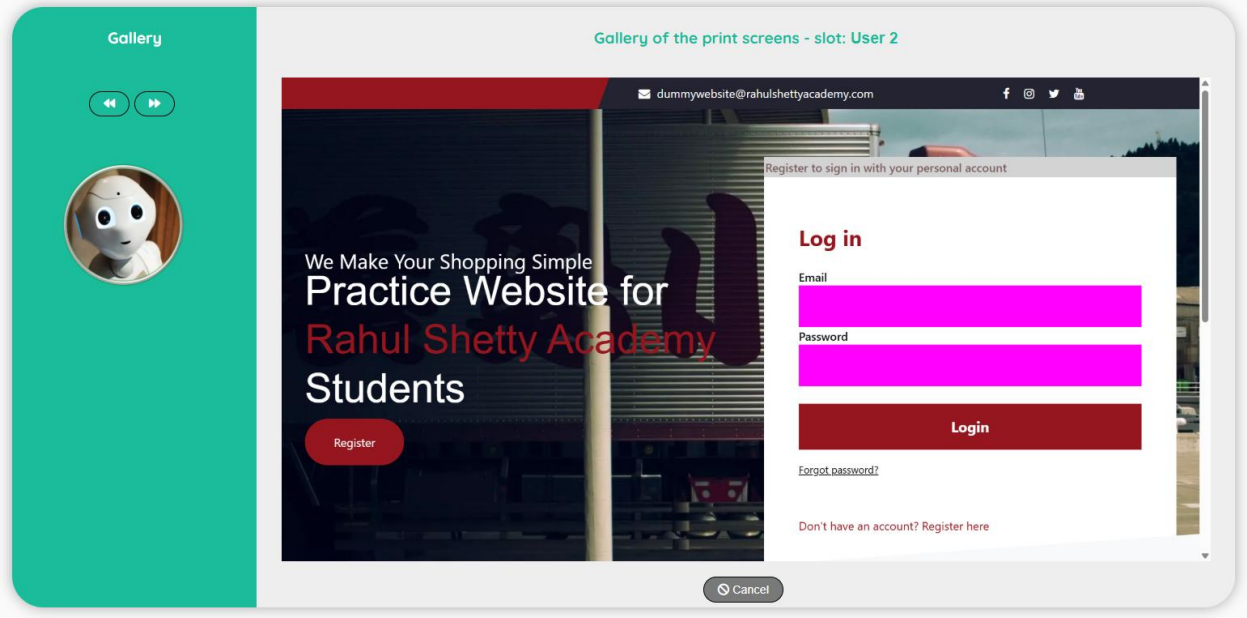
Baseline Name	Name the base line to use
Print screen Slot	From 1 to 5
Mask 1 → 5	You can add up to 5 locator to exclude for the print screen (you can use a word from the dictionary)

Example(s):

imageBaseline	Orders	1	Take a print screen and store the image as a baseline. The baseline is visible in the slot 1.

In this example, we used a mask on the login and the password.
You can see that the boundaries of the elements are replaced by a pink box.

The color is defined by Playwright and it's a color that will help for the comparison (used by the function imageDifference)



Function: imageDifference

Objectives

Method to take a print screen compare the image with a baseline

Parameter(s)

Baseline Name	Name the base line to use
Print screen Slot	From 1 to 5
Baseline Slot	Optional: From 1 to 5 to store the baseline image in a slot
Mask 1 → 5	You can add up to 5 locator to exclude for the print screen (you can use a word from the dictionary)

Example(s):

imageDifference	Orders	1	-1	Compare the baseline "Orders" with a fresh print screen (stored in the slot 1)
imageDifference	Orders	1	2	Compare the baseline "Orders" with a fresh print screen (stored in the slot 1) and store the baseline image in the slot 2

Note 1: if the baseline picture doesn't exit, the function create the image and stop.
You can also use the function imageBaseline() to create/refresh the baseline.

Note 2: There is a known bug and the images are not directly visible in the print screen slot. A refresh of the browser is required (with a login/password) to display the images.

Note 3: If you want accurate differences, be sure you use the same mask(s) as in the function imageBaseline.

Function: imageDifferenceData

Objectives

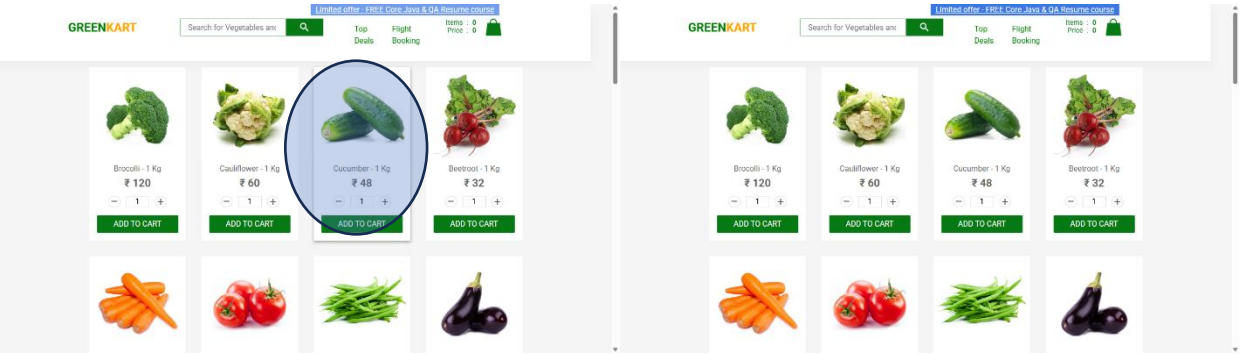
Method to store an attribute on an image difference in a variable after a call of the function imageDifference()

Parameter(s)

Parameter	Select a value in the list: Similarity, Difference, Different pixels, Image dimensions
Variable	Name of the variable

Example(s):

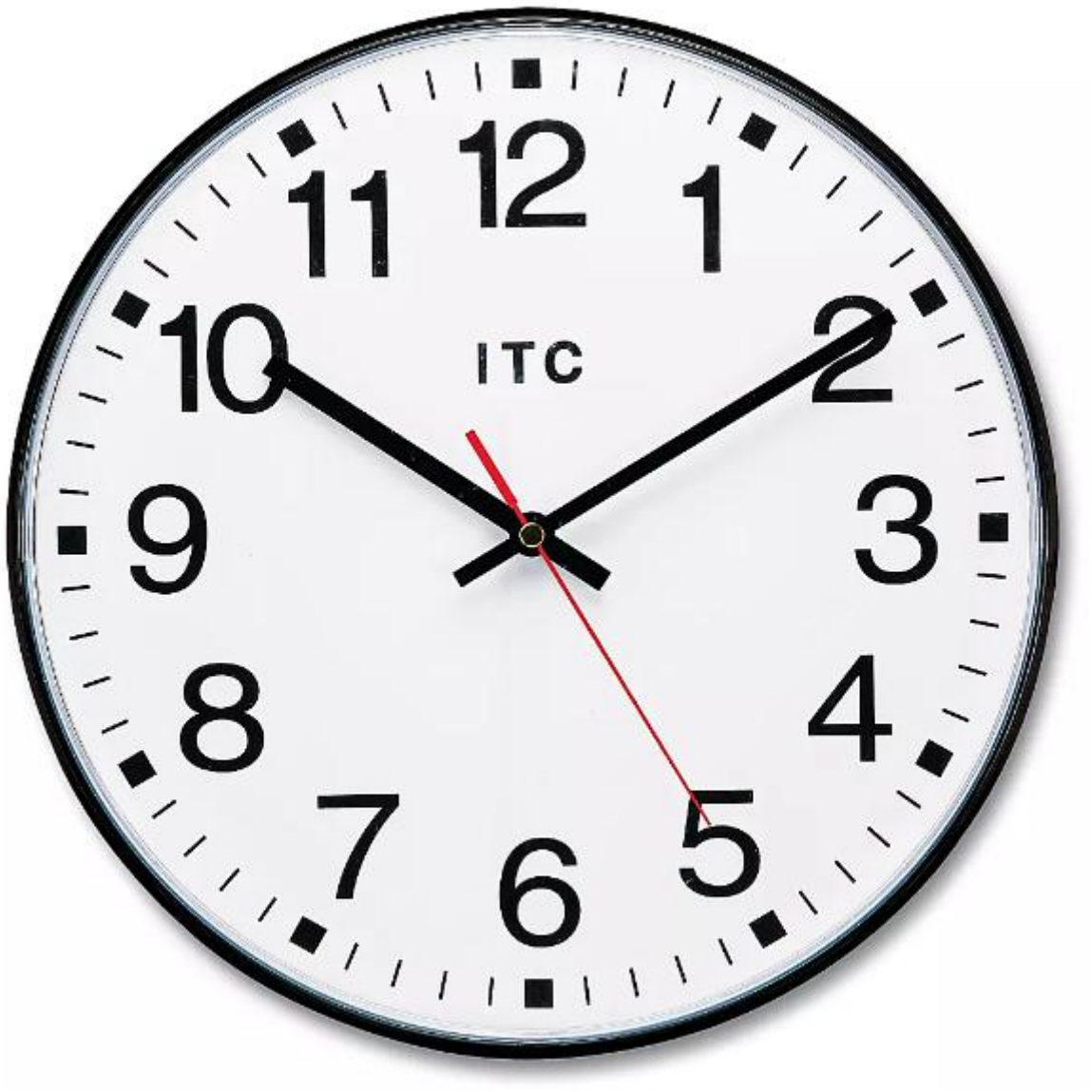
imageDifferenceData	Similarity	$Similarity	Similarity: 98.54%
imageDifferenceData	Difference	$Difference	Difference: 1.46%
imageDifferenceData	Different pixels	$Pixel	Different pixels: 13489 out of 921600
imageDifferenceData	Image dimensions	$Size	Image dimensions: 1280x720



In this example, the cucumbers are bigger on the left picture due to a mouse over.

Note: you can use the functions imageDifference() and imageDifferenceData() to detect a change in the user interface and send a message to the responsible of the documentation for instance.

Epoch Date Functions



Function: epoch

Objectives

Method to get a date converted into epoch (Unix) date and time.

Parameter(s)

Date	A date in any valid format
Format	Any valid format (E.g.: 'DD/MM/YYYY HH:mm:ss')
Variable	Name of the variable to store the result (starting with $)

Example(s):

epoch	22/05/2024, DD/MM/YYYY, $EpochDate	Convert a date

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #epoch: 22/05/2024, DD/MM/YYYY, §EpochDate

Function: epochDate

Objectives

Method to convert an epoch date into a date.

Parameter(s)

Epoch Date	An epoch date
Format	Any valid format (E.g.: 'DD/MM/YYYY HH:mm:ss')
Variable	Name of the variable to store the result (starting with $)

Example(s):

epochDate	$EpochDate, DD/MM/YYYY, $Date	Convert an epoch date

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #epochDate: $EpochDate, DD/MM/YYYY, §Date

Function: epochAddHour

Objectives

Method to get a date + hour(s) converted into epoch (Unix) date and time.

Parameter(s)

Date	A date in any valid format or NOW for the current date time
Format	Any valid format (E.g.: 'DD/MM/YYYY HH:mm:ss')
Hour	Number of hour(s) to add to the date
Variable	Name of the variable to store the result (starting with $)

Example(s):

epochAddHour	NOW, DD/MM/YYYY, 2, $Date	Add 2 hours and convert in epoch

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #epochAddHour: NOW, DD/MM/YYYY, 2, §Date

Function: epochAddMinute

Objectives

Method to get a date + minute(s) converted into epoch (Unix) date and time.

Parameter(s)

Date	A date in any valid format or NOW for the current date time
Format	Any valid format (E.g.: 'DD/MM/YYYY HH:mm:ss')
Minute	Number of minute(s) to add to the date
Variable	Name of the variable to store the result (starting with $)

Example(s):

epochAddMinute	NOW, DD/MM/YYYY, 10, $Date	Add 10 seconds and convert in epoch

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #epochAddMinute: NOW, DD/MM/YYYY, 10, §Date

Function: epochAddSecond

Objectives

Method to get a date + second(s) converted into epoch (Unix) date and time.

Parameter(s)

Date	A date in any valid format or NOW for the current date time
Format	Any valid format (E.g.: 'DD/MM/YYYY HH:mm:ss')
Second	Number of second(s) to add to the date
Variable	Name of the variable to store the result (starting with $)

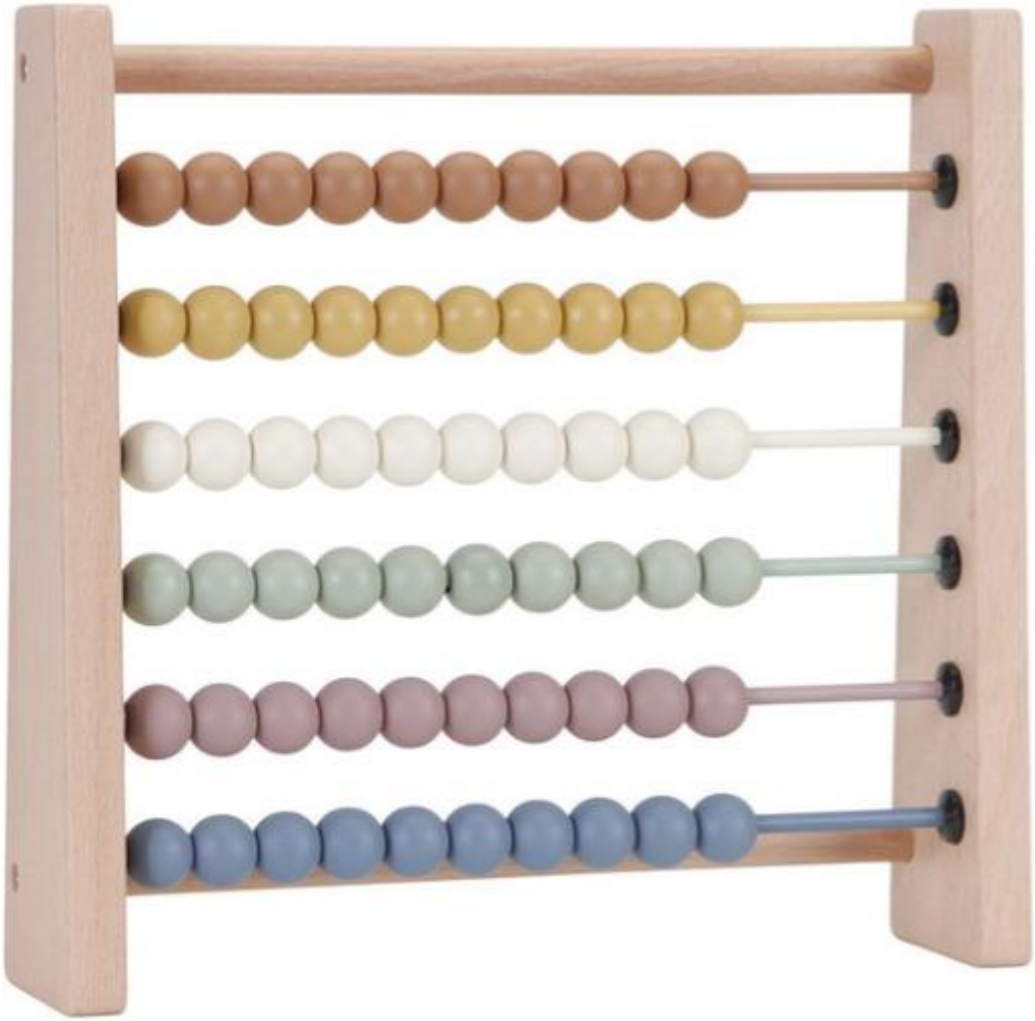
Example(s):

epochAddSecond	NOW, DD/MM/YYYY, 5, $Date	Add 5 minutes and convert in epoch

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #epochAddSecond: NOW, DD/MM/YYYY, 5, §Date

Table Functions



Function: getTableHeader

Objectives

Method to get a header from a table.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Row	The row position in the table
Column	The column position in the table
Variable	Name of the variable to store the result (starting with $)

Example(s):

getTableHeader	@APP_Amount, 1, 3, $AmountID	Get the value of the header (1,3)

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #getTableHeader: @APP_Amount, 1, 3, §AmountID

Function: getTableData

Objectives

Method to get a value from a cell of a table.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Row	The row position in the table
Column	The column position in the table
Variable	Name of the variable to store the result (starting with $)

Example(s):

getTableData	@APP_Amount, 1, 3, $Amount	Get the value of the cell (1,3)

Note: in the rule, the sign $ for the variable must be replaced by the sign: §

Example: #getTableData: @APP_Amount, 1, 3, §Amount

Function: setTableData

Objectives

Method to set a value into a cell of a table.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Row	The row position in the table
Column	The column position in the table
Value	Value or variable (starting with $)

Example(s):

setTableData	@APP_Amount, 1, 3, $Amount	set the amount into the cell (1,3)

Function: clickCell

Objectives

Method click on a cell of a table.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Row	The row position in the table
Column	The column position in the table
Delay	Duration (in second(s) after the click)

Example(s):

clickCell	@APP_Amount, 1, 3, 5	Click in the cell (1,3)

Function: countTableRow

Objectives

Method count the row(s) of a table.

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Variable	Name of the variable to store the result (starting with $)

Example(s):

countTableRow	@APP_Table, $Row	Count the number of row(s) of a table

Note: in the rule, the sign $ for the variable must be replaced by the sign: §
Example: #countTableRow: @APP_Table, §Row

Function: searchTableData

Objectives

Method search for a value in a table at a specific column.
Result is stored in the variable $Row (-1 if not found).

Parameter(s)

Name	Xpath or dictionary word (starting with @) of the element or $GUI
Column	The column position in the table
Search Value	Value to search in the table
Occurrence	Occurrence of the search (default 1)

Example(s):

searchTableData	@APP_Table, 3, Belgium, 1	Search for Belgium in a table

HTTP Request

Function: httpGet

Objectives

Perform a http request using the GET method.
The result will be stored in the httpdata with the code: “httpget”

Parameter(s)

Url	Link to the REST API (can be a variable or a word in the dictionary)
Token	Optional: can be N/A or a variable or a text or the name of a .p12 certificate
Key	Optional: only if a .p12 certificate is provided
Code	Code to store the http result (by default httpGet)

Example(s):

httpGet	Get all the licenses using REST API	http://localhost:5000/api/license	<N/A>	<N/A>
httpGet	API with a .p12 certificate	http://localhost:5000/api/license	$Token	$Key

Note 1: to get the result use the functions httpSearchKeyValue and httpKeyCount (see below)
Note 2: The status of the transaction is stored in the variable $HttpStatus
Note 3: the certificate .p12 must be uploaded on the server by the Administrator

Function: httpPost

Objectives

Perform a http request using the POST method.
The result will be stored in the httpdata with the code: “httppost”

Parameter(s)

Url	Link to the REST API (can be a variable or a word in the dictionary)
Body	JSON string to define the body parameters (can be a variable or a dataset)
Token	Optional: can be N/A or a variable or a text (or empty)
Key	Optional: only if a .p12 certificate is provided
Code	Code to store the http result (by default httpPost)

Example(s):

httpPost	Get a Natural word	@URL_API_Natural	{"code": "@Reject_the", "language": "EN", "active": 1}	<N/A>	<N/A>
httpPost	API with a .p12 certificate	@URL_API_Cert	{"code": "ABC"}	Cert.p12	$Key

Note 1: Use only double quote in the Body expression.
Note 2: The status of the transaction is stored in the variable $HttpStatus
Note 3: to get the result use the functions httpSearchKeyValue and httpKeyCount (see below)
Note 4: the certificate .p12 must be uploaded on the server by the Administrator

Function: httpPut

Objectives

Perform a http request using the PUT method.
The result will be stored in the httpdata with the code: “httpput”

Parameter(s)

Url	Link to the REST API (can be a variable or a word in the dictionary)
Body	JSON string to define the body parameters (can be a variable)
Token	Optional: can be N/A or a variable or a text (or empty)
Code	Code to store the http result (by default httpPut)

Example(s):

httpPut	Update a user	@URL_API_User	{"userId": 1, "lastname": "Dupond", "firstname": "Tom"}	N/A

Note 1: Use only double quote in the Body expression.
Note 2: The status of the transaction is stored in the variable $HttpStatus
Note 3: to get the result use the functions httpSearchKeyValue and httpKeyCount (see below)

Function: httpDelete

Objectives

Perform a http request using the DELETE method.
The result will be stored in the httpdata with the code: “httpdelete”

Parameter(s)

Url	Link to the REST API (can be a variable or a word in the dictionary)
Token	Optional: can be N/A or a variable or a text (or empty)
Code	Code to store the http result (by default httpDelete)

Example(s):

httpDelete	Delete a product	http://localhost:5000/api/product/10	$Token

The status of the transaction is stored in the variable $HttpStatus

Function: httpSearchKeyValue

Objectives

Get data from a previous httpGet, httpPost, httpPut, … or SAMLTransaction functions

Parameter(s)

Code	Code of the http data
Key	Key in the result (<ALL> for a dump of the result)
Key Occurrence	Position of the key in the result (1 by default)
Parent Key	Key of the parent (Granparent) in the result (<N/A> by default)
Parent Occurrence	Position of the parent (Granparent)in the result (1 by default)
Variable	Name of the variable to store the result (-1 if not found with $Error = 1) or a Boolean if a search value is used
Operation	Equal or Contains
Search value	(optional) <N/A> or a value to search in the result

Example(s) of httpSearchKeyValue:

Code	Key	Occ	Parent	Occ	Variable	Op	search	Comment
httpget	IDCard	1	<N/A>	1	$IDCard			No parent
httpget	IDCard	1	IDCards	2	$IDCard2			With parent
Httpget	<ALL>							Dump of the result in the log
Httpget	CardNr	1	IDCards	1	$Exist	contains	591286124036	True

Example, of the http data result

```
{"success":1,
 "data":
        <IDCards>
            <IDCard Type="195">
              <CardNr>591286124036</CardNr>
              <ExpiryDate><Century>20</Century><Year>16</Year><Month>03</Month><Day>02</Day></ExpiryDate>
            </IDCard>
            <IDCard Type="196">
              <CardNr>123286124036</CardNr>
              <ExpiryDate><Century>20</Century><Year>26</Year><Month>03</Month><Day>02</Day></ExpiryDate>
            </IDCard>
        </IDCards>
}
```

Function: httpKeyCount

Objectives

Count the number of occurrences of a key in the http result data

Parameter(s)

Code	Name of the element (or a variable)
Search key	Select a value in the list: Equal or Contains
Parent key	The value to search (default <N/A>)
Variable	Name of the variable to store the result (-1 if not found and $Error = 1)

Example(s):

httpKeyCount	httpget	FamilyMember	<N/A>	$httpCount	Return 4
httpKeyCount	httpget	Name	FamilyMembers	$httpCount	Return 4

In the example, the result of the httpdata is:

```
success: 1,
data:
 <FamilyMembers>
        <FamilyMember>
                <FamilyRole><Label>Epouse</Label></FamilyRole>
                <Name><LastName>Hubens</LastName><FirstName>Christianne</FirstName></Name>
        </FamilyMember>
        <FamilyMember>
                <FamilyRole><Label>Fille</Label></FamilyRole>
                <Name><LastName>Sermon</LastName><FirstName>Annelies</FirstName></Name>
        </FamilyMember>
        <FamilyMember>
                <FamilyRole><Label>Fils</Label></FamilyRole>
                <Name><LastName>Sermon</LastName><FirstName>Ronald</FirstName></Name>
        </FamilyMember>
        <FamilyMember>
                <FamilyRole><Label>Petit-fils</Label></FamilyRole>
                <Name><LastName>Dupuis</LastName><FirstName>Tim</FirstName></Name>
        </FamilyMember>
 </FamilyMembers>
```

Now, you can use the httpSearchKeyValue () to get all the items of the search via a Loop.
Put a condition: $HttpCount > 0 and set the occurrence of the key with $Loop

Function: SAML_Assertion

Objectives: Create a SAML Assertion

Parameter(s)

Certificate Dataset	Name of the dataset to use for the certificate
Assertion URL	URL for the assertion
Assertion Dataset	Name of the dataset to use for the assertion

Example(s):

SAML_Assertion	#SAML_Certificate	/v1.0/assertions/	#AB_Policy

Example of Certificate Dataset:

#SAML_Certificate_url	https://myserver.be	URL of the certificate
#SAML_Certificate_name	myCertificate.p12	Name of the certificate
#SAML_Certificate_key	Key123	Secret key of the certificate

Note 1: the certificate must be uploaded by the Administrator on the Robot server
Note 2: You can use the certificates: .p12, .pfx or .crt (with .key)
Note 3: It's very important to respect the data set code: _url, _name and _key (case sensitive), for the name of the dataset header, there is no restriction, you can use whatever you want!

Example of Assertion Dataset:

#AB_Policy_policy	AB_POLICY	Policy name
#AB_Policy_certificate	Pub1234cbg789…	Public key or <publickey>
#AB_Policy_requestId	<requestID>	Or enter your own request ID

Note 1: you need to check the requirements of your SAML assertion to define the correct dataset
Note 2: <publickey> (non sensitive) is not working with all the certificates, if you have an error, you have to provide manually the public key!
Note3: <requested> (non sensitive) is used to generate a request with the format: REQ-<date now>

Function: SAML_Transaction

Objectives: Create a SAML Transaction.
The result of the transaction will be stored in the http data with the code equal to the name provided for the assertion dataset.
To get the http data result use the functions httpSearchKeyValue and httpKeyCount

Parameter(s)

Certificate Dataset	Name of the dataset to use for the certificate
Assertion Dataset	Name of the dataset to use for the assertion
Transaction URL	URL for the transaction
Transaction Dataset	Name of the dataset to use for the transaction

Example(s):

SAML_Transaction	#SAML_Certificate	#AB_Policy	/v1.0/transactions/	#ZZ99

Example of Certificate Dataset:

#SAML_Certificate_url	https://myserver.be	URL of the certificate
#SAML_Certificate_name	myCertificate.p12	Name of the certificate
#SAML_Certificate_key	Key123	Secret key of the certificate

Note 1: the certificate must be uploaded by the Administrator on the Robot server
Note 2: You can use the certificates: .p12, .pfx or .crt (with .key)
Note 3: It's very important to respect the data set code: _url, _name and _key (case sensitive), for the name of the dataset header, there is no restriction, you can use whatever you want!

Example of Assertion Dataset:

#AB_Policy_policy	AB_POLICY	Policy name
#AB_Policy_certificate	Pub1234cbg789…	Public key or <publickey>
#AB_Policy_requestId	<requestID>	or enter your own request ID

Note 1: you need to check the requirements of your SAML assertion to define the correct dataset
Note 2: <publickey> (non sensitive) is not working with all the certificates, if you have an error, you have to provide manually the public key!
Note3: <requested> (non sensitive) is used to generate a request with the format: REQ-<date now>

Example of Transaction Dataset:

#ZZ99_txMessage	%ZZ99 ABCD00070123456	txMessage
#ZZ99_signed	00	signed
#ZZ99_requestId	<requestID>	or enter your own request ID

Note 1: you need to check the requirements of your SAML transaction to define the correct dataset
Note2: <requested> (non sensitive) is used to generate a request with the format: REQ-<date now>

Function: SOAP_postData

Objectives: Create a SOAP POST http request.
The result of the transaction will be stored in the http data with the code equal to the name provided for the assertion dataset.
To get the http data result use the functions httpSearchKeyValue and httpKeyCount

Parameter(s)

Soap URL	URL for the SOAP endpoint
Soap Dataset	Name of the dataset to use for the Soap request

Example(s):

SOAP_postData	http://server/services/PersonEndpoint	#getPerson

Example of Certificate Dataset:

#getPerson_soapenv	http://schemas.org/soap/envelope/	Soap envelope
#getPerson_core	http://www.server/core	Soap core
#getPerson_header	<!-- No header -->	No header in this example
#getPerson_body	<core:getPerson> <!-- person identifier --> <core:nationalNumber> 123456789 </core:nationalNumber> </core:getPerson>	Body specific for the request

Note: It's very important to respect the data set code: _soapenv, _core, _header and _body (case sensitive). If there is no header, just add a comment to avoid empty data!
For the name of the dataset header, there is no restriction, you can use whatever you want!

Technically, the information will be used like that:

```
const soapEnvelope = `
    <soapenv:Envelope xmlns:soapenv="${soapData.soapenv}"
                xmlns:core="${soapData.core}">
    <soapenv:Header>
       <core:dabsHeader>
          ${soapData.header}
       </core:dabsHeader>
    </soapenv:Header>
    <soapenv:Body>
          ${soapData.body}
    </soapenv:Body>
    </soapenv:Envelope>`;

  console.log('soapEnvelope', soapEnvelope)

  const response = await fetch(url, {
    method: 'POST',
    headers: {
      "Content-Type": "text/xml;charset=UTF-8",
      "SOAPAction": ""
    },
    body: soapEnvelope
  });
```

Advanced functions

Function: callScenario

Objectives

Method to execute a scenario based on its id

Parameter(s)

Scenario ID	Id of the scenario (visible in the detail of the scenario)

Example(s):

callScenario	126	Execute the tests of the scenario 126

Function: callSuite

Objectives

Method to execute a suite based on its id

Parameter(s)

Suite ID	Id of the suite (visible in the detail of the suite header)

Example(s):

callSuite	25	Execute the tests of the suite 25

Function: startTimer

Objectives

Start a timer to measure a performance

Parameter(s)

Topic	A short name to identify the timer

Example(s):

startTimer	login	Start a timer to measure the performance of the login

Function: stopTimer

Objectives

Stop a timer and store the elapsed time in the database.
Note: The timer is global to an application (not specific to a user)

Parameter(s)

Environment	Name of the environment
Topic	The identifier of the timer (must be the same name as in the startTimer)

Example(s):

stopTimer	PROD	login	Store the elapsed time in the database

At the end of the execution, a variable is automatically created with a rounded value (to the second(s)).
The name of the variable is $Timer<Topic> (topic with the first letter in upper case)
Example: $TimerLogin.
This variable can be used in the speak function to announce a text like that:
"Exceution successfully executed in $Timer<Topic> seconds"

Note: See also the chapter on the performance in the Designer manual

Function: promptAI

Objectives

Thanks to the library @zerostep, we can use a ChatGPT prompt to request something.
Unfortunately, this function is not working in a REST API environment to request testing operation like 'click on the submit button'

Parameter(s)

Prompt	A sentence to request something to chatGPT
Variable	Name of the variable to store the result

Example(s):

promptAI	Generate a realistic first name for a French people	$Name	ChatGPT prompt

Note: The library is free for only 500 requests per month (after, it's 40$ per month)

This function is experimental and cannot be used for production.
In June 2025: I sent a email to the @zerostep team to ask for a solution with a REST API.
Until now, no answer received!

Mobile functions

Function: phoneConnect

Objectives

Establish a connection with a mobile device

Parameter(s)

Note: Please, first, read how to configure Android Studio in the Design User Manual

Function: phoneTap

Objectives

Simulate the tap on the screen of a mobile device

Parameter(s)

Type:	Type of type: **text**: Text, **element**: Element, **web**: Web page
Element	Name of the element on the mobile (see also Mobile scan in the Designer User Manual)

Example(s):

phoneTap	text	Google	Tap on the Google icon on the screen

Note: The library is free for only 500 requests per month (after, it's 40$ per month)

Function: phoneFill

Objectives

Simulate the fill of an element on the phone

Parameter(s)

Type:	Type of type: **phone**: Phone, **web**: Web page
Element	Name of the element on the mobile (see also Mobile scan in the Designer User Manual)
Value	Value to enter

Example(s):

phoneFill	phone	com.android.chrome:id/url_bar	https://amazon.com
phoneFill	web	input[name="k"]	usb cable

Note: the phone and the web use two different methods to fill a value.
Be careful, the web element can be different from a webpage running on a desktop browser!
There is ,o **Enter** key at the end of the fill, so use the function phonePress() to do that

Function: phonePress

Objectives

Simulate the fill of an element on the phone

Parameter(s)

Type:	Type of type: **phone**: Phone, **web**: Web page
Element	Name of the element on the mobile (see also Mobile scan in the Designer User Manual)
Key	Enter, Home, Tab, Next, Previous, Home

Example(s):

phoneFill	phone	com.android.chrome:id/url_bar	Enter
phoneFill	phone	<N/A>	Home
phoneFill	web	input[name="k"]	Enter

Note: the phone and the web use two different methods to fill a value.
For the **Home** key on the phone, you don't need to specify a element

Function: phoneUrl

Objectives

Simulate open of a web page on the browser (Chrome)

Parameter(s)

Url:	Link to the web page

Example(s):

phoneUrl	https://amazon.com	Go to Amazon.com

Function: phoneCapture

Objectives

Method to take a print screen and store the image on a slot (up to 5 slots available)

Parameter(s)

Slot	From 1 to 5
Capture mode	**0**: Normal view, **1**: Full page

Example(s):

phoneCapture	1	0	Take a normal print screen and store it in the slot 1
phoneCapture	1	1	Take a full print screen and store it in the slot 1

ODBC Database functions

Function: ODBC_Generic

Objectives

Store a generic ODBC Connection string.
See also the Designer User Manual for the installation of the ODBC drivers.

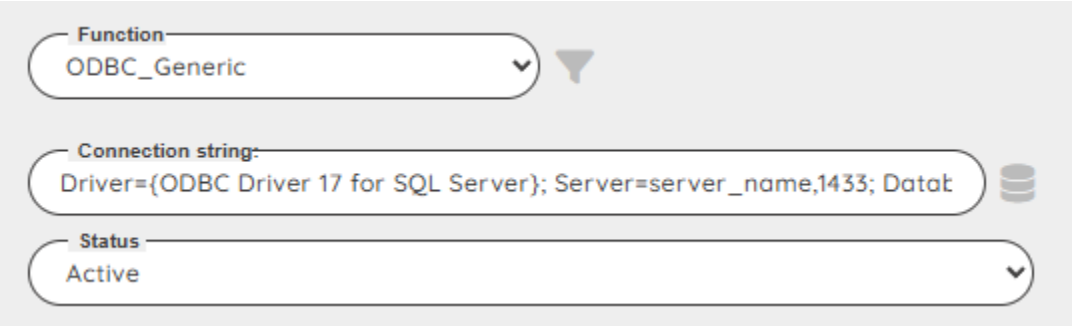
Parameter(s)

connectString	The connection string delimitied by ;

Example(s):

ODBC_Generic	Driver={ODBC Driver 17 for SQL Server}; Server=server_name,1433; Database=database_name; Uid=username; Pwd=password;	Example of connect string for SQL Server. For the clarity of the string, we added carriage returns but this must be omitted in the parameter

Note: you can use the function ODBC_Generic or a more specialized function like ODBC_SQLServer



Function: ODBC_SQLServer

Objectives

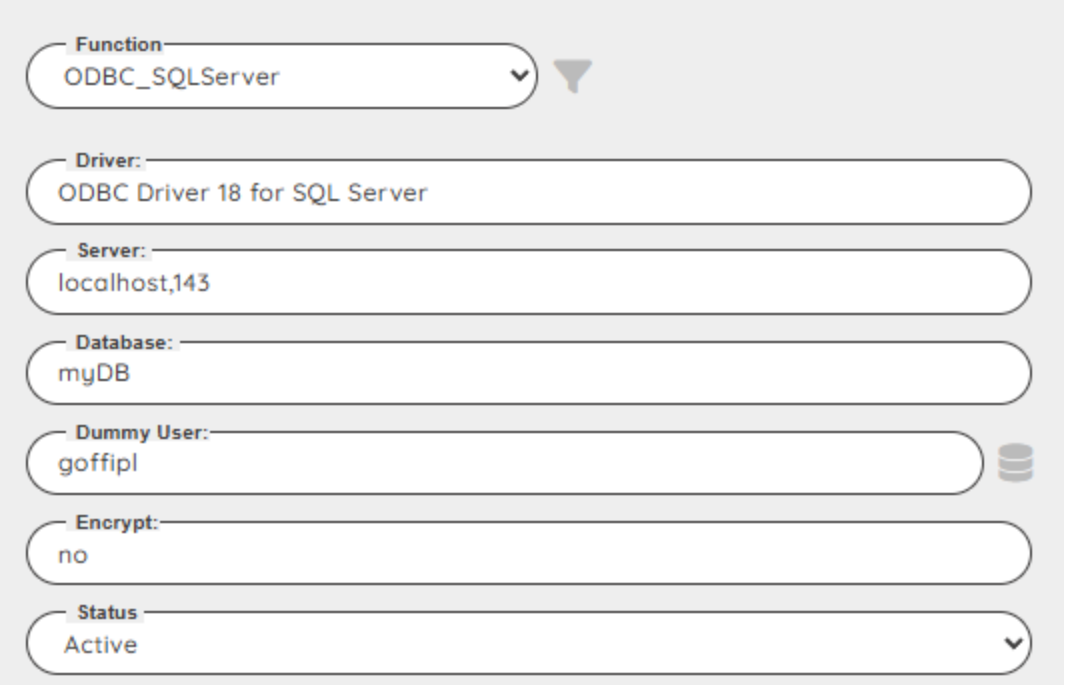
Store a SQL Server ODBC Connection string.
See also the Designer User Manual for the installation of the ODBC drivers.

Parameter(s)

Driver	Name of the driver (be sure that the default corresponds to your configuration)
Server	Name of the server
Database	Name of the database
DummyUser	Name of the dummy user
Encrypy	yes / no (by default)

Example(s):

ODBC_SQLServer	Driver=ODBC Driver 18 for SQL Server Server=server_name,1433 Database=database_name DummyUser=Phil Encrypt=no	Example of connect string for SQL Server. DummyUser must be provided to get the login and the password (crypted or not) Note: dummy user can be <ME>



Function: ODBC_Oracle

Objectives

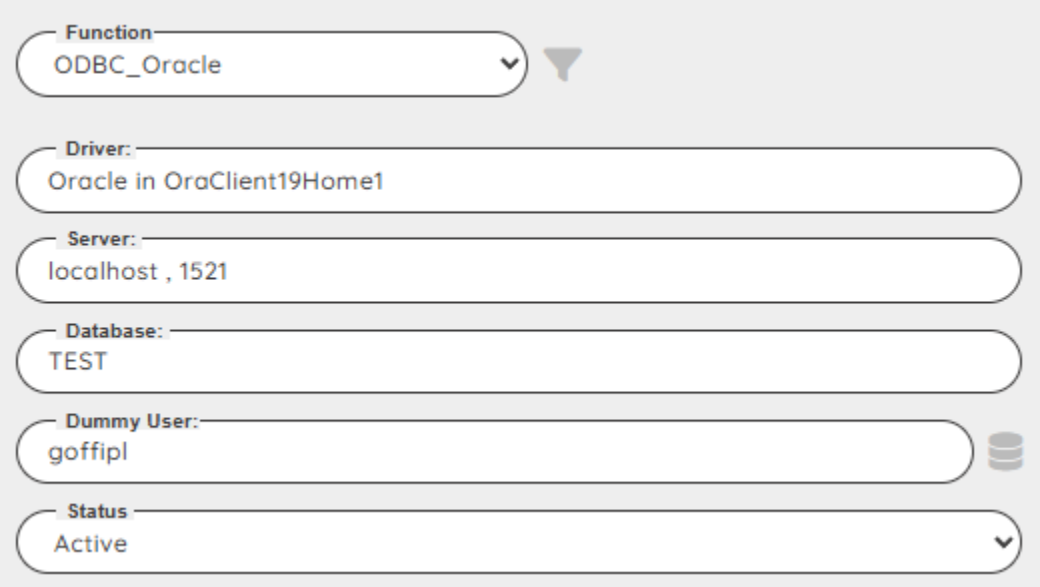
Store an Oracle ODBC Connection string.
See also the Designer User Manual for the installation of the ODBC drivers.

Parameter(s)

Driver	Name of the driver (be sure that the default corresponds to your configuration)
Server	Name of the server
Database	Name of the database
DummyUser	Name of the dummy user

Example(s):

ODBC_Oracle	Driver=Oracle in OraClient19Home1 Server=server_name,1433 Database=database_name DummyUser=Phil	Example of connect string for Oracle. DummyUser must be provided to get the login and the password (crypted or not) Note: dummy user can be <ME>



Function: ODBC_MySQL

Objectives

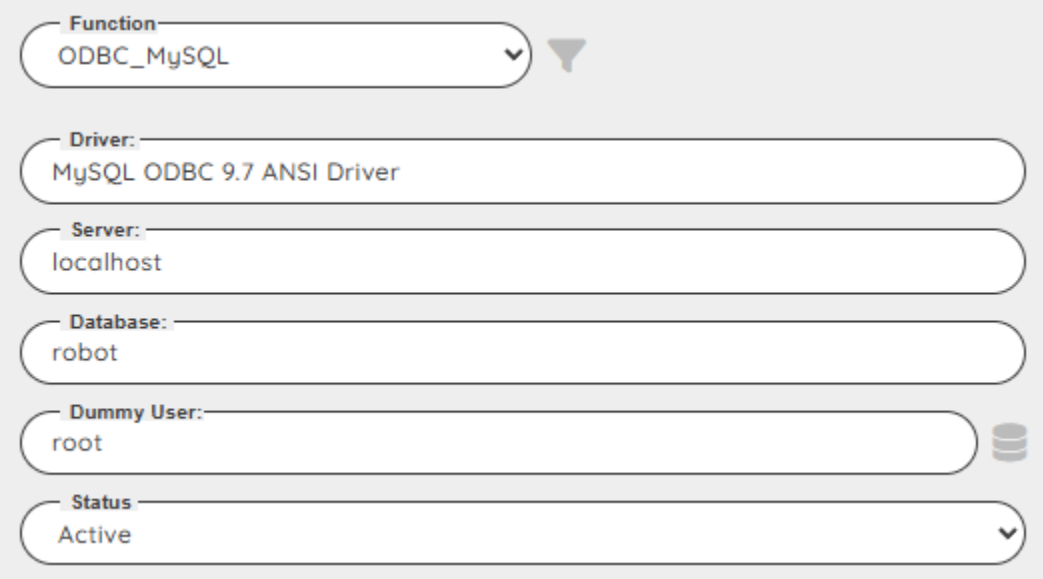
Store a MySQL ODBC Connection string.
See also the Designer User Manual for the installation of the ODBC drivers.

Parameter(s)

Driver	Name of the driver (be sure that the default corresponds to your configuration)
Server	Name of the server
Database	Name of the database
DummyUser	Name of the dummy user

Example(s):

ODBC_MySQL	Driver= MySQL ODBC 9.7 ANSI Driver Server=localhost Database=database_name DummyUser=root	Example of connect string for MySQL. DummyUser must be provided to get the login and the password (crypted or not) Note: dummy user can be <ME>



Function: ODBC_ExecuteSQL

Objectives

Execute a SQL statement with an ODBC Connection
Prior to the execution, you need to define first a connection string with the database.
The function can be used for a select as well as for DML operation (insert, delete…)

Parameter(s)

Driver	Name of the driver (be sure that the default corresponds to your configuration)
SQL	Name of the server with ? for each parameter
Parameters	Value of the parameter(s) separated by ;

Example(s):

ODBC_ExecuteSQL	SQL: SELECT * FROM `user` WHERE login = ? Parameter: goffipl	Select
ODBC_ExecuteSQL	SQL: INSERT INTO `todo` (`todo`, `position`) VALUES (?, ?) Parameter: Test;4	DML Insert
ODBC_ExecuteSQL	SQL: DELETE FROM `todo` WHERE todo= ? Parameter: Test	DML Delete

Note:

- The variable $Error is set to 1 in case of error (otherwise 0)
- The variable $ODBCRow contains the number of rows (Select) or the affected rows (DML)
- In case of a select, the result is stored in an internal storage (Robot database) that can be used by the function ODBC_Datavalue().
  This information is persistent by user, so you can share the data between scenarios.

Function: ODBC_DataCount

Objectives

Extract the number of a record of a previous SELECT.
Prior to the execution, you need to execute a Select statement.

Parameter(s)

Variable	Name of the variable to store the number of records (starts with $)

Example(s):

ODBC_DataCount	$Count	Store the number of records of the last select statement into the variable $Count

Note: as the select information is persistent by user, you can use the function in different scenarios.

Function: ODBC_DataValue

Objectives

Extract data of a column of a previous SELECT.
Prior to the extract, you need to execute a Select statement.

Parameter(s)

Column	Name of the column to extract the data
Row	Row of the record (> 0 and <= DataCount)
variable	Name of the variable to store the data (starts with $)

Example(s):

ODBC_DataValue	Column: lastName Row: 1 Variable: $Value	Store the last name of the first record into the variable $Value

Note:

- The variable $Error is set to 1 in case of error (otherwise 0)
- as the select information is persistent by user, you can use the function in different scenarios.
- If the result is undefined, check the name of the column. The variable $Error will be set but no fatal error with be throw!